

BEE EXAM MASTER NOTES – 42 IMPORTANT QUESTIONS

Q1. Ohm's Law

Definition: Current is directly proportional to voltage. Formula: $V=IR$. Key Points: Temperature constant, current increases with voltage.

Q2. KCL & KVL

KCL: $\sum I=0$. KVL: $\sum V=0$. KCL based on conservation of charge, KVL based on conservation of energy.

Q3. Resistor, Inductor, Capacitor

Resistor opposes current, Inductor opposes current change, Capacitor opposes voltage change.

Q4. KCL/KVL Numericals

Apply current balance in nodes and voltage balance in loops.

Q5. Series & Parallel Circuits

Series: $R_{eq}=R_1+R_2+R_3$. Parallel: $1/R_{eq}=1/R_1+1/R_2$.

Q6. Branch Current

Use Ohm's law $I=V/R$.

Q7. RC Circuit

$X_C=1/2\pi fC$. Current leads voltage.

Q8. RL Circuit

$X_L=2\pi fL$. Current lags voltage.

Q9. RLC Circuit

$Z=\sqrt{R^2+(X_L-X_C)^2}$. Resonance when $X_L=X_C$.

Q10. Power Triangle

$P=VI\cos\phi$, $Q=VI\sin\phi$, $S=VI$, $PF=\cos\phi=P/S$.

Q11. Impedance Equation

$Z=\sqrt{R^2+(X_L-X_C)^2}$.

Q12. RLC Numericals

$I=V/Z$ and $\tan\phi=(X_L-X_C)/R$.

Q13. DC Generator Principle

Works on Faraday's law of electromagnetic induction.

Q14. DC Generator Construction

Main parts: yoke, poles, armature, commutator, brushes.

Q15. DC Generator EMF

$E_g=(P\phi ZN)/(60A)$.

Q16. DC Motor

Converts electrical energy into mechanical energy.

Q17. Torque Equation

Torque proportional to flux and armature current. $T\propto\phi I_a$.

Q18. Back EMF

$E_b=V-I_aR_a$. Controls armature current and speed.

Q19. Rotating Magnetic Field

Produced by 3-phase AC supply.

Q20. Single Phase IM

Uses stator and squirrel cage rotor.

Q21. Why IM Not Self Starting

Starting torque is zero due to equal opposite torques.

Q22. Synchronous Speed

$N_s = 120f/P$.

Q23. Transformer Principle

Works on mutual induction.

Q24. Transformer Construction

Core, primary winding, secondary winding.

Q25. Transformer EMF Equation

$E = 4.44fN\phi_m$.

Q26. Transformer Losses

Hysteresis, eddy current, copper losses.

Q27. Ideal Transformer

No losses, no leakage flux, 100% efficiency.

Q28. Transformer Ratio

$V_1/V_2 = N_1/N_2$.

Q29. MCB MCCB ELCB

MCB-domestic, MCCB-industrial, ELCB-shock protection.

Q30. Pipe & Plate Earthing

Pipe earthing is economical, plate earthing for industries.

Q31. Switchgear

Includes fuse, relay, MCB, MCCB, isolator.

Q32. SFU

Switch + Fuse for isolation and protection.

Q33. Earthing Methods

Pipe, plate, rod, strip earthing.

Q34. Circuit Breakers

Automatically disconnect faulty circuits.

Q35. Back EMF Significance

Protects motor and regulates speed.

Q36. Resonance

At resonance $X_L = X_C$, current maximum.

Q37. Loss Reduction

Use laminated core and silicon steel.

Q38. Power Factor

$\cos\phi = P/S$. High PF improves efficiency.

Q39. Transformer Advantages

High efficiency, low maintenance.

Q40. DC Motor Advantages

High starting torque, good speed control.

Q41. Synchronous Speed Equation

$N_s = 120f/P$.

Q42. Single Phase IM Not Self Starting

Net starting torque = 0.

MOST IMPORTANT FORMULAS

Important Formula	Use
$V = IR$	Ohm's Law
$\Sigma I = 0$	KCL
$\Sigma V = 0$	KVL
$X_L = 2\pi fL$	Inductive Reactance
$X_C = 1/(2\pi fC)$	Capacitive Reactance
$Z = \sqrt{R^2 + (X_L - X_C)^2}$	RLC Impedance
$f_r = 1/(2\pi\sqrt{LC})$	Resonance Frequency
$P = VI\cos\phi$	Real Power
$Q = VI\sin\phi$	Reactive Power
$S = VI$	Apparent Power
$N_s = 120f/P$	Synchronous Speed
$E_g = (P\phi ZN)/(60A)$	DC Generator EMF
$E = 4.44fN\phi_m$	Transformer EMF